

Filippo Ficarra

Education

- Sep 2022 – **MSc in Computer Science, ETH Zürich**
- Nov 2024 Major in Machine Intelligence and minor in Information Security. Relevant Courses: Large Language Models, Natural Language Processing, Computer Vision, Computational Intelligence Lab, Advanced Machine Learning, Probabilistic AI, Applied Cryptography and Information Security Lab.
- Master Thesis at Rycolab: Investigated how LLMs learn words versus humans measuring word acquisition using surprisal-based metrics. Accepted at **NAACL 2025**. [A Distributional Perspective on Word Learning in Neural Language Models](#). 📄
 - Semester Project at Rycolab: Analyzed how varying vision-to-text ratios influences language learning in FLAVA multimodal models. *How Multimodality Influences Word Acquisition: Do Images Help Language Models Mimic Human Behavior?*
 - SwissAI in collaboration with CSCS and Rycolab. Contributed to the training data and model architecture team for a large-scale multimodal model.
- Sep 2019 – **BEng in Computer Engineering, Politecnico di Torino, Final Grade: 110/110 cum laude**
- Jul 2022 Completed my Bachelor's degree fully funded by "Percorso Giovani Talenti" excellence track. Relevant courses: Data Structures and Algorithms, Object-Oriented Programming.

Experience

- Jan 2025 – **Research Scientist Intern, IBM Research Zürich**
- July 2025 Developed advanced machine learning models to optimize and accelerate laboratory procedures.
- Collaborated with chemists to train a multimodal Transformer that predicts molecular structures from mixture IR spectra and molecular formulas, achieving 34% Top-1 accuracy, surpassing human experts and representing the first approach to this task. Accepted as a workshop paper at **NeurIPS 2025**. [Language Model Enabled Structure Prediction from Infrared Spectra of Mixtures](#). 📄
 - IBM Granite for Single Atom Catalysts: Built a knowledge extraction system using LangChain and Qwen2.5, and instruction-tuned IBM Granite 3.2 8B to generate SAC synthesis procedures.
 - Large-Scale LLM Training: Trained Llama3 (8B–405B) models on 2–2048 Nvidia GH200 GPUs with DeepSpeed ZeRO-3 in a slurm cluster, securing a major research grant.
- Sep 2023 – **Teaching Assistant, ETH Zürich**
- Sep 2024 Assisted course staff in designing and testing assignments, creating teaching materials for *Large Language Models* and *Information Security Lab*, and led lab sessions to teach core concepts and guide students through practical problem-solving.
- Mar 2022 – **Backend Engineer Intern, ORBYTA TECH**
- Jun 2022 Designed and optimized a scalable REST API backend for a B2B clothing platform using C# with the ABP Framework and MySQL, delivering efficient and reliable data services.
- Nov 2021 – **IT Consultant, JEToP**
- Aug 2022 Built the front-end of a cross-device podcast application using React Native, TypeScript, and Tailwind CSS, implementing multimedia playback, dynamic routing, and interactive pages to enhance user experience; collaborated with UI/UX designers to refine interfaces, client-ready mobile app

Skills

AI	Deep Learning, NLP, Large Language Models, Computer Vision	Languages	Python, JavaScript, C#, Java, C, SQL, Bash
Python Ecosystem	PyTorch, Deepspeed, HuggingFace, Sklearn, LangChain, FastAPI	Technologies	Docker, slurm, GIT, Google Cloud, IBM Cloud

References

- [1] Filippo Ficarra, Marvin Alberts, and Teodoro Laino. Language model enabled structure prediction from infrared spectra of mixtures. In *AI for Accelerated Materials Design - NeurIPS 2025*, 2025.
- [2] Filippo Ficarra, Ryan Cotterell, and Alex Warstadt. A distributional perspective on word learning in neural language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11184–11207, Albuquerque, New Mexico, April 2025.